

EV applications.

The production process of silicon carbide devices is distinct and more refined compared to traditional silicon, which helps in achieving higher quality and performance. Manufacturers have optimised production lines specifically for SiC, enhancing the overall quality of the final product. This attention to detail in manufacturing ensures that SiC devices are not only more efficient but also more reliable over their operational lifespan.

When selecting the optimal packaging for power devices, one critical question arises: What package will best maintain device efficiency and longevity? The 20247 package, commonly used for SiC devices, is an exemplary choice. It offers excellent low-loss performance and superior thermal resistance, which are essential attributes for sustaining the high efficiency and longevity of the devices. This particular packaging choice is well-suited to meet the rigorous demands of high-power applications, such as EV charging and solar energy systems, where both efficiency and reliability are paramount.

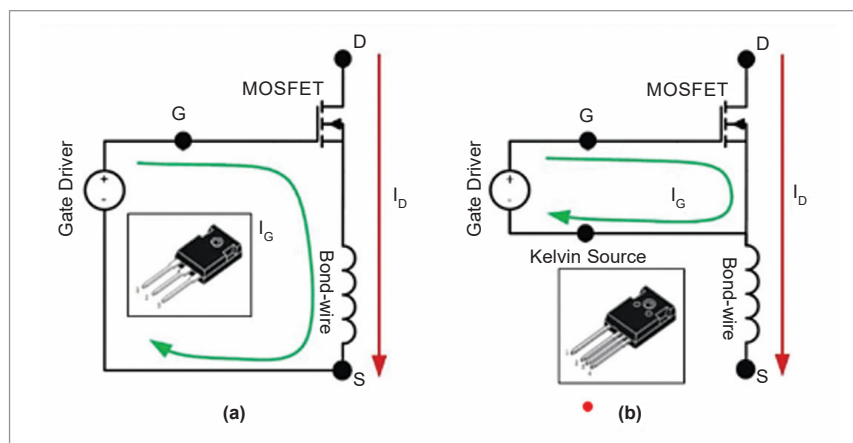
Role of SiC in EV performance enhancement

Silicon carbide enhances the performance of electric vehicles in several key areas:

Faster charging times. SiC's high efficiency and thermal conductivity contribute to faster and more reliable charging, reducing downtime and enhancing user convenience.

Improved thermal management. The ability to handle higher temperatures allows SiC devices to operate more efficiently under stress, prolonging their lifespan and reducing the likelihood of overheating.

Compact system design. The high system frequency potential of SiC makes it possible to design



Advantage of Kelvin-Source at SiC MOSFET (Credit: Diotec Semiconductor)

smaller and lighter components, which is vital for optimising the space and weight of EVs.

The ongoing developments in SiC technology are focused on expanding its application beyond just high-power settings to include a broader range of electronic components. These innovations are paving the way for SiC to play a more dominant role in the future of EVs and other high-demand electronics, offering a blend of performance and environmental benefits.

Addressing industry challenges and misconceptions

Despite its numerous benefits, the integration of SiC technology comes with challenges, particularly around its relative novelty in the market. As companies push the boundaries of SiC applications, they also work to dispel common misconceptions about its complexity and cost. The potential for SiC to reduce operational and maintenance costs substantially outweighs the initial investment, making it an increasingly attractive option for high-power applications.

The future of SiC in EV technology looks promising with ongoing research focused on improving durability and performance under extreme conditions. Innovations in

packaging and integration techniques continue to emerge, paving the way for more sophisticated applications not only in automotive but also in renewable energy systems and industrial power solutions.

As the automotive industry continues to evolve, the role of silicon carbide in advancing electric vehicle technology becomes increasingly crucial. Its ability to drive efficiency, reduce costs, and improve performance positions SiC as a cornerstone of future developments in EV tech. For those looking to stay ahead in the fast-paced world of automotive innovations, embracing SiC technology offers a competitive edge that will define the next generation of electric vehicles. The practical application of SiC in EVs has demonstrated significant improvements in charging infrastructure and onboard power systems. Case studies and real-world applications, available through resources like IBS Electronics, showcase how SiC technology enhances the charging speed and efficiency of electric vehicles, contributing to the industry's shift towards more sustainable mobility solutions. **EFY**

This article is based on a webinar hosted by Diotec and IBS Electronics. It has been transcribed and curated by Akanksha Sondhi Gaur, Research Analyst and Technical Journalist at EFY.